

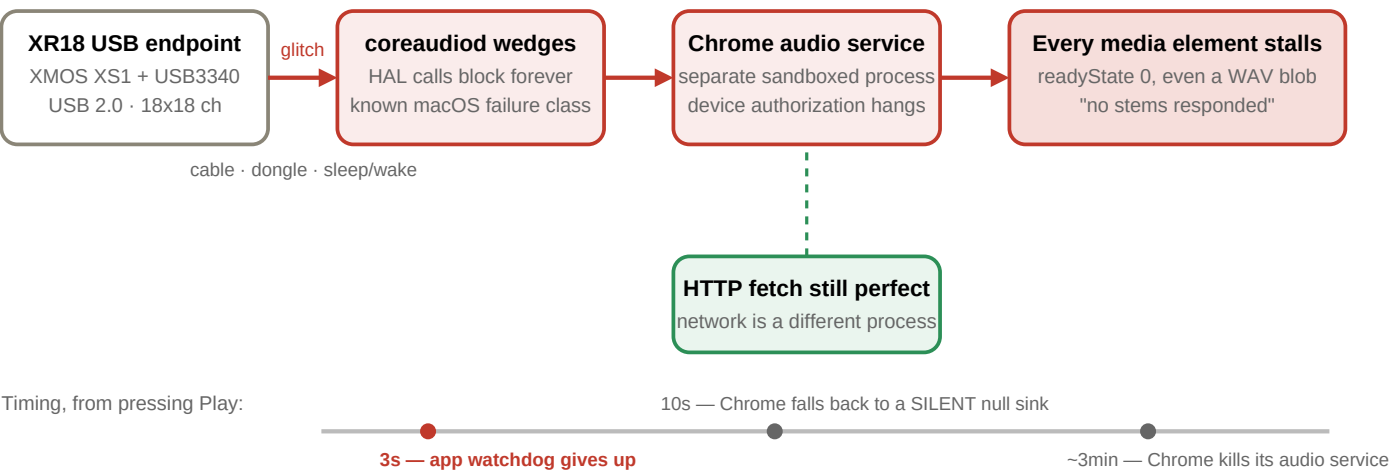
Postmortem: The Audio Wedges

simpleStem Performer rig · researched and written 2026-07-04 · three wedges July 2–4 · full citations in docs/10_AUDIO_WEDGE_DEEP_DIVE.md

What happened

Three times in two days, every song in the portal failed with “**no stems responded after 3s**” while the server, network, and files were provably healthy (HTTP fetches of the same stems streamed at full speed; even an in-memory WAV blob would not load). Once, `sudo killall coreaudiod` fixed it in 4 ms. Once, only physically replugging the XR18’s USB cable fixed it. Once, a `coreaudiod` kick left the XR18 re-enumerated as a 2-channel device (“orphaned”) before it recovered to 18 channels on its own.

The failure chain



Chrome runs all CoreAudio calls in a separate sandboxed **audio service process** — which is why networking stays perfect while audio dies. New media elements wait on **device authorization** from that service; when `coreaudiod` is hung the wait runs to Chrome’s 10-second timeout, after which playback proceeds **silently** into a null sink. `simpleStem`’s watchdog fires at 3 seconds — inside that window — so a wedged audio device always presents as “no stems responded.”

Two diseases, one symptom

Incident	Wedge side	What fixed it	Why
Jul 3, evening	Daemon-side (macOS)	<code>killall coreaudiod</code> — audio back in 4 ms	Classic <code>coreaudiod</code> wedge; documented macOS failure class, heaviest on Apple Silicon, sleep/wake the top trigger
Jul 4, morning	Board-side (XR18)	USB unplug/replug only; <code>killall</code> did NOT stick	The board’s XMOS USB controller hung; a fresh daemon kept re-opening a dead endpoint. Only a physical reset clears it
2-ch “orphaned” state	Re-enumeration	Recovered on its own / replug	Matches Mac reports that the X AIR 2-in/2-out USB mode is broken — re-enumeration can land there

The recovery ladder — what each rung resets

Step	Resets	Right when	Gotcha
1 · Replug XR18 USB (5 s out)	The board's XMOS USB controller	Board-side hang — killall didn't stick	Needs hands at the board
2 · <code>sudo killall coreaudiod</code>	The macOS audio daemon	Daemon-side wedge	May re-enumerate XR18 as 2-ch; <code>launchctl kickstart</code> is blocked since macOS 14.4 — use <code>killall</code>
3 · Reload the portal tab	Chrome's client audio state	Always, after rungs 1–2	Clients hold corrupted state across a daemon bounce (documented)
4 · Power-cycle the XR18 (10 s)	Everything board-side	Replug didn't do it	Re-pick the device in macOS Sound

Prevention, ranked by evidence strength

Fix	Evidence	Notes
One quality USB-C→USB-B cable, ≤2 m, straight into the Mac — no dongle, no passive hub	STRONG — multiple confirmed community fixes	Our 15-foot (4.6 m) cable is towards the practical limit for USB 2.0 audio and the prime suspect for recurring endpoint glitches. Shorter cable ordered 2026-07-04.
XR18 firmware current (USB fixes shipped in 1.23)	VENDOR — Behringer changelog	Rig updated to 1.25 on 2026-07-04. Changelog for 1.23: “Fixed: USB audio issues (mostly at 44K1)”.
USB mode 18-in/18-out, 48 kHz pinned everywhere	GOOD	The 2-in/2-out mode is reported broken on Mac; 44.1 kHz is where the firmware bugs lived. simpleStem is all-48k by policy.
No sleep at gigs: AC power, lid open, caffeineate	MODERATE	Sleep/wake is the most-cited coreaudiod trigger. performer.sh now runs caffeineate -dis automatically for the life of the rig (2026-07-04). The hot-corner screensaver still works — caffeineate blocks sleep, not a manually invoked saver.
Wi-Fi radio OFF at the venue	MODERATE	Removes mDNS churn, Drive-sync retries, captive-portal probing; forces link-local addressing. XR18 control rides Ethernet, the portal is localhost, stems are in local cache — nothing on stage needs the radio. New WIFI pill in the portal header toggles it (2026-07-04).
If a hub is unavoidable: externally powered Thunderbolt dock	MODERATE	The only hub class that resolved the MacBook USB-2 disconnect wave.
Smart hub + uhubctl for scripted port power-cycling	NICHE	The only software substitute for a physical replug on macOS.

Assessment of the standing hypotheses

“**Known issue with Mac Core Audio**” — **CONFIRMED**. coreaudiod wedging with USB interfaces is a recurring, vendor-acknowledged macOS failure class (Apple Dev Forums threads through Sonoma and Sequoia; Apple shipped USB/audio fixes in 14.4.1; Focusrite and Rogue Amoeba both document it). “**The XR18 gives up and shuts down the channel**” — **PARTLY**. Nothing suggests the firmware shuts the port down deliberately; the evidence points at the USB link glitching and the board's XMOS controller then hanging in a state only a physical reset clears. The distinction matters: the cure is a better link (short cable, no dongle, current firmware), not different board settings.

Follow-up implemented in the app: mode-aware transport pills, single-lane visualizer, Wi-Fi pill, automatic caffeineate. Candidate next step: when the stem watchdog fires, auto-run the WAV-blob probe and show “AUDIO DEVICE WEDGED — first-aid ladder” instead of “pick another song.”

Sources: Apple Dev Forums 742465 / 748228 · Apple Communities 255788454, 255046431 · Apple 14.4.1 release notes · Focusrite KBs · Behringer X18/XR18 firmware changelog · behringer.world · Cockos & Steinberg forums · vogelchr XR18 teardown · Chromium source (audio_device_factory, audio_renderer_impl, audio_thread_hang_monitor, services/audio) · crbug 160920 / 615589 / 422522 · webrtc 4799 · Hancke (2017) · uhubctl. Full annotated list: docs/10_AUDIO_WEDGE_DEEP_DIVE.md.